COMPUTER NETWORKS LAB – PRACTICAL EXAM CHEAT SHEET

212CSE3302 | Cisco Packet Tracer | Experiments 5–12

Format for every experiment:

1. AIM
2. PROCEDURE
3. CONFIGURATION / COMMANDS
4. VERIFICATION / OUTPUT
5. RESULT

The commands are separated clearly so you can directly practise them in Cisco Packet Tracer.

EXPERIMENT 5 – INTRA-VLAN CONFIGURATION

1. AIM

To design and implement an Intra-VLAN network using Cisco Packet Tracer by creating VLANs, assigning switch ports to appropriate VLANs, configuring trunk links and verifying communication between devices belonging to the same VLAN.

2. PROCEDURE

1. Open Cisco Packet Tracer.
2. Place two switches and the required PCs and laptop.
3. Connect PCs to switches using Copper Straight-Through cables.
4. Connect both switches using a Copper Cross-Over cable or Auto-Connect.
5. Configure IP addresses and subnet masks for all PCs.
6. Open the switch CLI and enter privileged configuration mode.
7. Create VLAN 10 and VLAN 20.
8. Assign the required switch ports to their respective VLANs.
9. Configure the connection between the switches as a trunk port.
10. Verify VLAN configuration and test connectivity using the ping command.

Konfigurasi VLAN Trunk - Cisco Packet Tracer - Pintar Network
pintar-network.com

3. IP ADDRESS CONFIGURATION

Device	IP address	Subnet mask

PC0	192.168.10.1	255.255.255.0
PC1	192.168.10.2	255.255.255.0
PC2	192.168.20.2	255.255.255.0
PC3	192.168.20.3	255.255.255.0
PC4	192.168.10.3	255.255.255.0
PC5	192.168.20.5	255.255.255.0
PC6	192.168.20.4	255.255.255.0
Laptop0	192.168.10.4	255.255.255.0

4. CONFIGURATION COMMANDS

A. Switch 0 – Create VLANs

```
enable
configure terminal

vlan 10
exit

vlan 20
exit
```

B. Assign ports to VLANs

```
interface fa0/1
switchport mode access
switchport access vlan 10
exit

interface fa0/2
switchport mode access
switchport access vlan 20
exit
```

Repeat the port assignments according to the connected PCs.

C. Configure trunk port

```
interface fa0/3
switchport mode trunk
exit
```

Perform the corresponding VLAN and trunk configuration on Switch 1.

D. Verify configuration

```
show vlan brief
show interfaces trunk
```

5. VERIFICATION / OUTPUT

On PC0 → Desktop → Command Prompt:

```
ping 192.168.10.2
ping 192.168.10.3
```

Expected output: Successful ping replies between devices in the same VLAN.

6. RESULT

Thus, the Intra-VLAN network was successfully designed and configured using Cisco Packet Tracer, and communication between devices belonging to the same VLAN was verified.

EXPERIMENT 6(a) – INTER-VLAN USING LAYER 3 SWITCH

1. AIM

To design and implement Inter-VLAN routing using a Layer 3 switch by creating VLANs, configuring Switch Virtual Interfaces (SVIs), enabling IP routing and establishing communication between different VLANs.

2. PROCEDURE

1. Open Cisco Packet Tracer.
2. Add PCs, Layer 2 switches and one multilayer (Layer 3) switch.
3. Connect the PCs to the switches.
4. Connect the switches to the multilayer switch.
5. Assign IP addresses to all PCs.
6. Create VLAN 10 and VLAN 20 on the switches.
7. Assign switch ports to the respective VLANs.
8. Configure trunk ports between the switches.
9. Enable IP routing on the multilayer switch.
10. Create SVI interfaces for VLAN 10 and VLAN 20.
11. Assign gateway IP addresses to the SVIs.
12. Verify connectivity between different VLANs using ping.

Cisco Layer 3 Switch InterVLAN Routing Configuration - Study CCNA
study-ccna.com

3. IP ADDRESS CONFIGURATION

Device	IP address	Default gateway
PC0	192.168.10.11	192.168.10.100
PC1	192.168.10.12	192.168.10.100
PC2	192.168.20.11	192.168.20.100

PC3 192.168.20.12 192.168.20.100

Subnet mask: `255.255.255.0`

4. CONFIGURATION COMMANDS

A. Create VLANs

```
enable
configure terminal

vlan 10
exit

vlan 20
exit
```

B. Assign access ports

```
interface fa0/1
switchport mode access
switchport access vlan 10
exit

interface fa0/2
switchport mode access
switchport access vlan 20
exit
```

C. Configure trunk port

```
interface fa0/3
switchport trunk encapsulation dot1q
switchport mode trunk
exit
```

If the switch does not support the encapsulation command, omit that line and configure trunk mode directly.

D. Enable Layer 3 routing

```
ip routing
```

E. Configure VLAN 10 gateway

```
interface vlan 10
ip address 192.168.10.100 255.255.255.0
no shutdown
exit
```

F. Configure VLAN 20 gateway

```
interface vlan 20
ip address 192.168.20.100 255.255.255.0
```

```
no shutdown
exit
```

5. VERIFICATION / OUTPUT

```
show vlan brief
show ip interface brief
show ip route
```

From PC0:

```
ping 192.168.20.11
```

Expected output: Successful communication between VLAN 10 and VLAN 20.

6. RESULT

Thus, Inter-VLAN routing using a Layer 3 switch was successfully configured and communication between different VLANs was verified.

EXPERIMENT 6(b) – INTER-VLAN USING ROUTER-ON-A-STICK

1. AIM

To design and implement Inter-VLAN routing using the Router-on-a-Stick method by creating VLANs, configuring trunk links, creating router subinterfaces and enabling communication between different VLANs.

2. PROCEDURE

1. Open Cisco Packet Tracer.
2. Add one router, one or more switches and PCs.
3. Connect PCs to the switches.
4. Connect the switch to the router using a trunk link.
5. Assign IP addresses to all PCs.
6. Create VLAN 10, VLAN 20 and VLAN 30 on the switches.
7. Assign switch ports to the respective VLANs.
8. Configure the switch port connected to the router as a trunk port.
9. Create router subinterfaces for each VLAN.
10. Configure 802.1Q encapsulation using the `encapsulation dot1Q` command.
11. Assign gateway IP addresses to the router subinterfaces.
12. Enable the physical router interface.
13. Test communication between VLANs using ping.

ROTEAMENTO INTER-VLAN - Cisco Community
community.cisco.com

3. IP ADDRESS CONFIGURATION

Device	IP address	Default gateway
PC0	192.168.10.10	192.168.10.100
PC1	192.168.20.10	192.168.20.100
PC2	192.168.30.10	192.168.30.100
PC3	192.168.10.11	192.168.10.100
PC4	192.168.20.11	192.168.20.100
PC5	192.168.30.11	192.168.30.100

Subnet mask: `255.255.255.0`

4. CONFIGURATION COMMANDS

A. Create VLANs on the switch

```
enable
configure terminal

vlan 10
vlan 20
vlan 30
exit
```

B. Assign access ports

```
interface fa0/1
switchport mode access
switchport access vlan 10
exit

interface fa0/2
switchport mode access
switchport access vlan 20
exit

interface fa0/3
switchport mode access
switchport access vlan 30
exit
```

C. Configure trunk port

Use the switch interface connected to the router. For example:

```
interface fa0/4
switchport mode trunk
exit
```

D. Configure router physical interface

```
enable
configure terminal

interface gigabitEthernet 0/0
no shutdown
exit
```

E. Configure VLAN 10 subinterface

```
interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.100 255.255.255.0
exit
```

F. Configure VLAN 20 subinterface

```
interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.100 255.255.255.0
exit
```

G. Configure VLAN 30 subinterface

```
interface gigabitEthernet 0/0.30
encapsulation dot1Q 30
ip address 192.168.30.100 255.255.255.0
exit
```

5. VERIFICATION / OUTPUT

```
show ip interface brief
show ip route
```

From PC0:

```
ping 192.168.20.10
ping 192.168.30.10
```

Expected output: Successful communication between all configured VLANs.

6. RESULT

Thus, Inter-VLAN routing using the Router-on-a-Stick method was successfully configured, and communication between different VLANs was verified.

EXPERIMENT 9 – DHCP CONFIGURATION

1. AIM

To design and implement DHCP configuration using Cisco Packet Tracer to automatically assign IP addresses, subnet masks and default gateway information to computers in a network.

2. PROCEDURE

1. Open Cisco Packet Tracer.
2. Add one router, one switch and the required PCs.
3. Connect the router to the switch.
4. Connect all PCs to the switch.
5. Open the router CLI.
6. Configure the router interface with an IP address.
7. Enable the router interface using `no shutdown`.
8. Exclude the IP addresses that should not be dynamically assigned.
9. Create a DHCP pool.
10. Configure the network address, default gateway and DNS server.
11. Open each PC and select Desktop → IP Configuration.
12. Select DHCP to obtain an IP address automatically.
13. Verify the assigned addresses using `ipconfig` and the router's DHCP commands.

[Configure Static Ip Packet Tracer at Jessica Dowie blog]
[storage.googleapis.com]

3. IP ADDRESS CONFIGURATION

Parameter	Value
Network address	192.168.1.0
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
DNS server	192.168.1.10
Excluded addresses	192.168.1.1 – 192.168.1.49

4. CONFIGURATION COMMANDS

A. Configure router interface

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

B. Exclude IP addresses

```
ip dhcp excluded-address 192.168.1.1 192.168.1.49
```

C. Create DHCP pool

```
ip dhcp pool LAN
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 192.168.1.10
exit
```

D. Verify DHCP configuration

```
show ip dhcp binding
show ip dhcp pool
show ip interface brief
```

5. PC CONFIGURATION

1. Click PC0.
2. Select Desktop.
3. Open IP Configuration.
4. Select DHCP.
5. Check the automatically assigned IP address.

Repeat for other PCs.

Command Prompt:

```
ipconfig
```

6. RESULT

Thus, DHCP was successfully configured using Cisco Packet Tracer, and IP addresses were automatically assigned to the client computers.

EXPERIMENT 10 – DNS CONFIGURATION / DNS PACKET ANALYSIS

1. AIM

To study and analyze DNS query and response packets using Wireshark and understand how domain names are resolved into IP addresses.

2. PROCEDURE

1. Open Wireshark.
2. Select the active network interface.
3. Start capturing network packets.
4. Open Command Prompt.
5. Execute the `ping` command using a domain name.
6. Allow Wireshark to capture the DNS query and response packets.

7. Apply the DNS display filter.
8. Select a DNS query packet.
9. Examine the DNS transaction ID, flags and questions.
10. Select the DNS response packet.
11. Examine the response flags and answer records.
12. Compare the DNS query and response packets.

winapi - How to intercept DNS queries in Windows - Stack Overflow
stackoverflow.com

3. COMMANDS

A. Display IP configuration

```
ipconfig /all
```

B. Generate DNS traffic

```
ping google.com
```

C. Find DNS information

```
nslookup google.com
```

D. Wireshark display filter

```
dns
```

4. IMPORTANT DNS PACKET DETAILS

Parameter	Description
Protocol	DNS
Port number	53
Query	Request sent by client
Response	Answer sent by DNS server
Transaction ID	Identifies DNS request and response
Flags	Indicates query or response details
Questions	Number of DNS questions
Answer RRs	Number of DNS answer records

Important difference:

- DNS Query: The client requests the IP address of a domain.
- DNS Response: The DNS server returns the corresponding IP address.

5. RESULT

Thus, DNS query and response packets were successfully captured and analyzed using Wireshark, and the process of domain name resolution was studied.

EXPERIMENT 11 – FTP SERVER CONFIGURATION

1. AIM

To design and implement an FTP server using Cisco Packet Tracer to transfer files between client computers and a server using the File Transfer Protocol.

2. PROCEDURE

1. Open Cisco Packet Tracer.
2. Add one router, one switch, one server and two PCs.
3. Connect all devices using appropriate cables.
4. Configure IP addresses, subnet masks and default gateways.
5. Wait until the network connections become active.
6. Open the Server.
7. Select Services → FTP.
8. Turn ON the FTP service.
9. Create a username and password.
10. Click Add to create the FTP user account.
11. Open PC0 → Desktop → Command Prompt.
12. Test connectivity to the FTP server using the ping command.
13. Enter the FTP server IP address using the `ftp` command.
14. Enter the configured username and password.
15. Use the `put` command to upload a file.
16. Use the `get` command to download a file.
17. Verify file transfer using the `dir` command.

How To Configure FTP Server in Cisco Packet tracer - Netizzan
netizzan.com

3. IP ADDRESS CONFIGURATION

The manual gives the following client configuration:

Device	IP address	Subnet mask	Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1

The FTP file-transfer example in the manual uses server IP `10.10.10.2`.

Important: Use the server IP and router interfaces matching your actual Packet Tracer topology. The client and server must have working IP connectivity, including routing if they are in different subnets.

4. SERVER CONFIGURATION

1. Click Server.
2. Select Services.
3. Select FTP.
4. Turn FTP service ON.
5. Enter the username and password.

Example:

Username	Password
student	123

6. Click Add.

5. CLIENT CONFIGURATION COMMANDS

A. Check IP address

```
ipconfig
```

B. Test server connectivity

```
ping 10.10.10.2
```

C. Connect to FTP server

```
ftp 10.10.10.2
```

Enter the username and password configured on the server.

D. Display available files

```
dir
```

E. Upload a file

```
put test.txt
```

F. Download a file

```
get test.txt
```

G. Exit FTP

```
quit
```

6. IMPORTANT FTP COMMANDS

Command	Function
`ftp`	Connect to FTP server
`put`	Upload file
`get`	Download file
`dir`	Display file list
`quit`	Exit FTP session
`pwd`	Display current directory

Protocol: FTP

Default ports: TCP 20 (traditional data connection), TCP 21 (control connection).

7. RESULT

Thus, the FTP server was successfully configured using Cisco Packet Tracer, and file uploading and downloading between the client and server were verified.

EXPERIMENT 12 – EMAIL SERVER CONFIGURATION

1. AIM

To design and implement an Email Server using Cisco Packet Tracer by configuring SMTP and POP3 services and establishing email communication between two client computers.

2. PROCEDURE

1. Open Cisco Packet Tracer.
2. Add one router, one switch, one server, one PC and one laptop.
3. Connect all devices using appropriate cables.
4. Configure the IP addresses, subnet masks and default gateways.
5. Configure the router interfaces for communication between the client and server networks.
6. Open Server → Desktop → IP Configuration.
7. Assign the server IP address and default gateway.
8. Open Server → Services → EMAIL.
9. Turn ON SMTP and POP3 services.
10. Enter the email domain name.
11. Create email accounts with usernames and passwords.
12. Open PC0 → Desktop → Email.
13. Configure the email address, incoming mail server and outgoing mail server.
14. Configure the laptop email settings using the second email account.
15. Compose an email from PC0 and send it to the laptop.

16. Open the laptop email application and click Receive.
17. Open the received email and send a reply.
18. Verify successful email communication.

Mail Server Create By Cisco Packet Tracer | PDF
slideshare.net

3. IP ADDRESS CONFIGURATION

According to your lab manual:

Device	IP address	Subnet mask	Default gateway
Router Fa0/0	192.168.1.1	255.255.255.0	—
Server	192.168.2.2	255.255.255.0	192.168.2.1
PC0	192.168.1.2	255.255.255.0	192.168.1.1
Laptop0	192.168.1.3	255.255.255.0	192.168.1.1

4. ROUTER CONFIGURATION COMMANDS

The server's gateway is `192.168.2.1`. Therefore, configure the router's second interface with that address.

A. Configure first interface

```
enable
configure terminal

interface fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

B. Configure second interface

```
interface fa0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

C. Verify router interfaces

```
show ip interface brief
```

5. EMAIL SERVER CONFIGURATION

A. Configure Server IP

Click Server → Desktop → IP Configuration.

```
IP Address:      192.168.2.2
```

```
Subnet Mask:     255.255.255.0
Default Gateway: 192.168.2.1
```

B. Enable Email Services

1. Click Server.
2. Select Services.
3. Select EMAIL.
4. Turn ON SMTP.
5. Turn ON POP3.
6. Enter the domain name:

```
example.com
```

7. Create the email accounts.

C. Create Email Users

Username	Password	Email address
pc0	123	pc0@example.com
laptop0	123	laptop0@example.com

Click Add after entering each account.

6. PC0 EMAIL CONFIGURATION

Click PC0 → Desktop → Email.

Enter the following details:

```
Your Name: PC0
Email Address: pc0@example.com
Incoming Mail Server: 192.168.2.2
Outgoing Mail Server: 192.168.2.2
Username: pc0
Password: 123
```

Click Save.

7. LAPTOP EMAIL CONFIGURATION

Click Laptop0 → Desktop → Email.

Enter:

```
Your Name: Laptop
Email Address: laptop0@example.com
Incoming Mail Server: 192.168.2.2
Outgoing Mail Server: 192.168.2.2
Username: laptop0
```

```
Password: 123
```

Click Save.

8. SEND AND RECEIVE EMAIL

From PC0:

1. Open Email.
2. Click Compose.
3. Enter recipient:

```
laptop0@example.com
```

4. Enter the subject and message.
5. Click Send.

From Laptop0:

1. Open Email.
2. Click Receive.
3. Open the received message.
4. Click Reply.
5. Send a reply to PC0.

9. IMPORTANT EMAIL PROTOCOLS

Protocol	Purpose	Port
SMTP	Sending emails	25
POP3	Receiving emails	110
IMAP	Email synchronization	143

10. RESULT

Thus, the Email Server was successfully configured using Cisco Packet Tracer by enabling SMTP and POP3 services, creating email accounts and verifying email communication between PC0 and Laptop0.

FINAL QUICK REVISION – IMPORTANT COMMANDS

Memorize these commands before entering the practical examination.

Experiment	Most important command
Intra-VLAN	`switchport access vlan 10`

Inter-VLAN (L3)	`ip routing`
Inter-VLAN (Router)	`encapsulation dot1Q 10`
DHCP	`ip dhcp pool LAN`
DNS	`nslookup google.com`
FTP	`ftp 10.10.10.2`
Email	SMTP ON + POP3 ON

Commonly Used Cisco Commands

```
enable
configure terminal
interface fa0/1
no shutdown
exit
show vlan brief
show interfaces trunk
show ip interface brief
show ip route
show running-config
```

Common PC Commands

```
ipconfig
ipconfig /all
ping 192.168.1.1
nslookup google.com
ftp 10.10.10.2
tracert google.com
```

You're right. Experiments 7 and 8 were missing from the cheat sheet. Add the following sections between Inter-VLAN and DHCP. I've kept the same format: AIM → PROCEDURE → IP CONFIGURATION → COMMANDS → VERIFICATION → RESULT, based on your Computer Networks Lab manual.

Experiment 7 – Checking Layer 2 Functionality Using Packet Tracer

AIM:
To check Layer 2 functionality using Cisco Packet Tracer by configuring network devices and verifying connectivity between PCs and laptops through a switch.

Devices Required:

- 1 Switch
- 4 PCs
- 2 Laptops
- Copper straight-through cables

IP Address Configuration:

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
Laptop0	192.168.1.2	255.255.255.0
PC1	192.168.1.3	255.255.255.0
PC2	192.168.1.4	255.255.255.0
Laptop1	192.168.1.5	255.255.255.0
PC3	192.168.1.6	255.255.255.0

PROCEDURE:

1. Open Cisco Packet Tracer.
2. Place one switch, four PCs and two laptops in the workspace.
3. Connect all end devices to the switch using copper straight-through cables.
4. Configure the IP addresses and subnet masks as shown above.
5. Wait until the link indicators turn green.
6. Open the Command Prompt on each PC.
7. Use `ipconfig` to verify the IP configuration.
8. Use `ping` to test communication between the devices.
9. Observe the switch's MAC address learning table.

COMMANDS:

```
ipconfig
ping 192.168.1.2
ping 192.168.1.3
ping 192.168.1.4
ping 192.168.1.5
ping 192.168.1.6
```

Switch Verification Command:

```
enable
show mac address-table
```

OUTPUT / VERIFICATION:

- Successful ping replies indicate connectivity between devices.
- The switch learns the MAC addresses of connected devices.
- MAC addresses are associated with the corresponding switch ports.

RESULT:
Thus, Layer 2 functionality was checked successfully using Cisco Packet Tracer by verifying device connectivity and MAC address learning.

Complete - 212CSE3302 - Computer Networks Lab manual (1)-2.pdf

Experiment 8 – Exploring Link State and Distance Vector Routing Protocols

This experiment has two parts: 8(a) Link State Routing and 8(b) Distance Vector Routing.

Experiment 8(a) – Link State Routing

AIM:
To design and implement Link State Routing using Cisco Packet Tracer.

Devices Required:

- Routers
- Switches
- PCs
- Serial and Ethernet cables

PROCEDURE:

1. Open Cisco Packet Tracer.
2. Place the required routers, switches and PCs.
3. Connect the devices according to the network topology.
4. Configure IP addresses on the PCs and router interfaces.
5. Enable the router interfaces using `no shutdown`.
6. Configure OSPF as the link state routing protocol.
7. Advertise the connected networks using OSPF.
8. Verify the routing table and test connectivity using ping.

ROUTER CONFIGURATION – OSPF (Example):

Use the following as a sample for a router connected to the `192.168.1.0/24` and `192.168.2.0/24` networks. Adapt the network statements to your actual topology.

```
enable
configure terminal

router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0

exit
```

VERIFICATION COMMANDS:

```
show ip route
show ip ospf neighbor
ping <destination-IP>
```

OUTPUT / VERIFICATION:

- OSPF discovers routes through neighboring routers.
- The routing table displays learned OSPF routes, marked with `O`.
- Successful ping confirms connectivity between networks.

RESULT:
Thus, Link State Routing was implemented successfully using OSPF in Cisco Packet Tracer.

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Complete - 212CSE3302 - Computer Networks Lab manual (1)-2.pdf

Experiment 8(b) – Distance Vector Routing

AIM:
To design and implement Distance Vector Routing using Cisco Packet Tracer.

Devices Required:

- Routers
- Switches
- PCs
- Serial and Ethernet cables

IP Address Configuration (as listed in the manual):

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0

PROCEDURE:

1. Open Cisco Packet Tracer.
2. Place the required routers, switches and PCs.
3. Connect the devices according to the topology.
4. Configure IP addresses on PCs and router interfaces.
5. Enable all router interfaces using `no shutdown`.
6. Configure RIP as the Distance Vector Routing protocol.
7. Advertise the connected networks using RIP.
8. Verify the routing table.
9. Test connectivity between different networks using ping.

ROUTER CONFIGURATION – RIP (Example):

Apply the appropriate network statements to each router according to its connected networks.

```
enable
configure terminal

router rip
version 2
no auto-summary
network 192.168.1.0
network 192.168.2.0

exit
```

VERIFICATION COMMANDS:

```
show ip route
show ip protocols
ping <destination-IP>
```

OUTPUT / VERIFICATION:

- Routers exchange routing information with their neighbouring routers.
- The routing table displays RIP-learned routes, marked with `R`.
- RIP selects routes based on hop count.
- Successful ping confirms communication between networks.

RESULT:
Thus, Distance Vector Routing was implemented successfully using RIP in Cisco Packet Tracer.

Complete - 212CSE3302 - Computer Networks Lab manual (1)-2.pdf
Complete - 212CSE3302 - Computer Networks Lab manual (1)-2.pdf